Week 3 Problem Set

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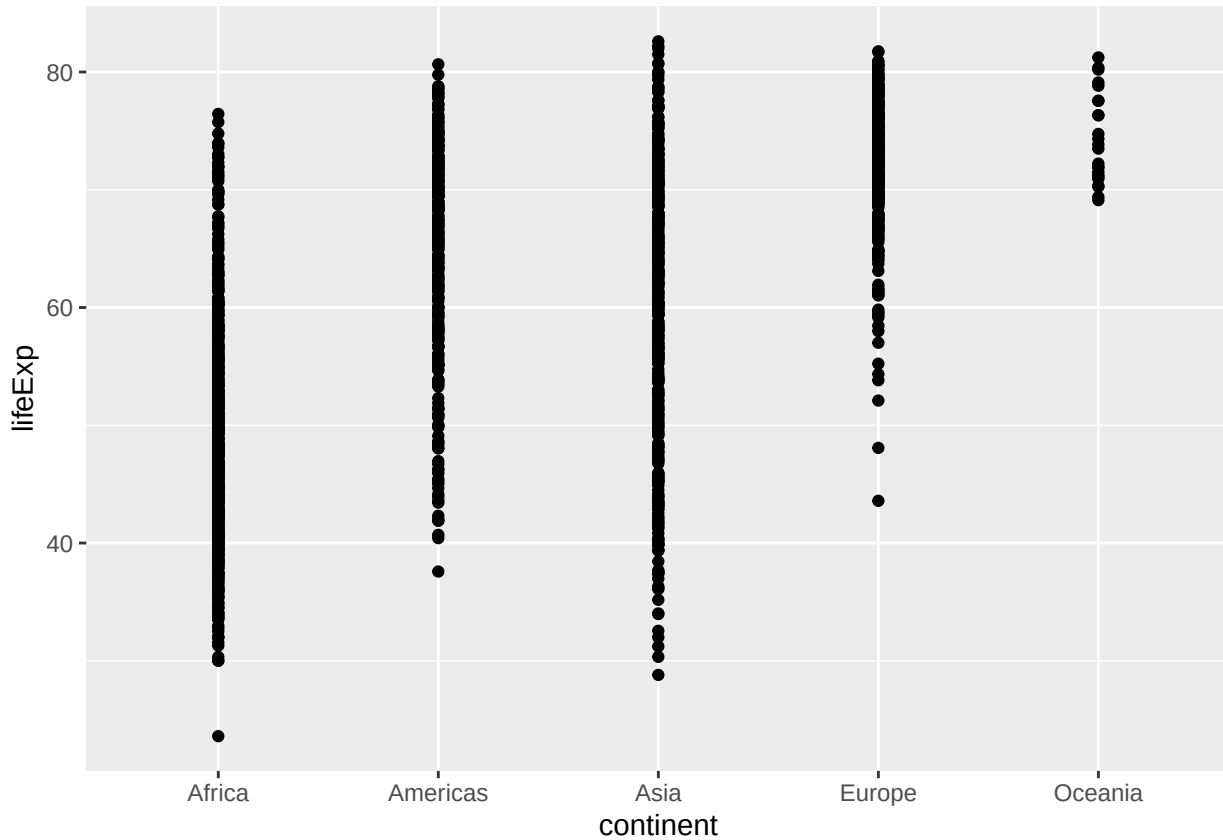
1. Start a new document, problemset3.Rmd, in your soc10problemsets repository. (You can download a template here, but be sure to save it in your own soc10problemsets repo). Change your name in the header. Load the `gapminder` data using `library(gapminder)`. Create a simple plot with two of the variables. **Commit** your changes.

Here I created a plot that illustrates life expectancy across continents. Meaning, the x axis is the regions and y axis is the average life expectancy for each continent Each point represents a country's life expectancy within a continent.

```
library(gapminder)
library(ggplot2)
```

```
## Warning: package 'ggplot2' was built under R version 4.5.2
```

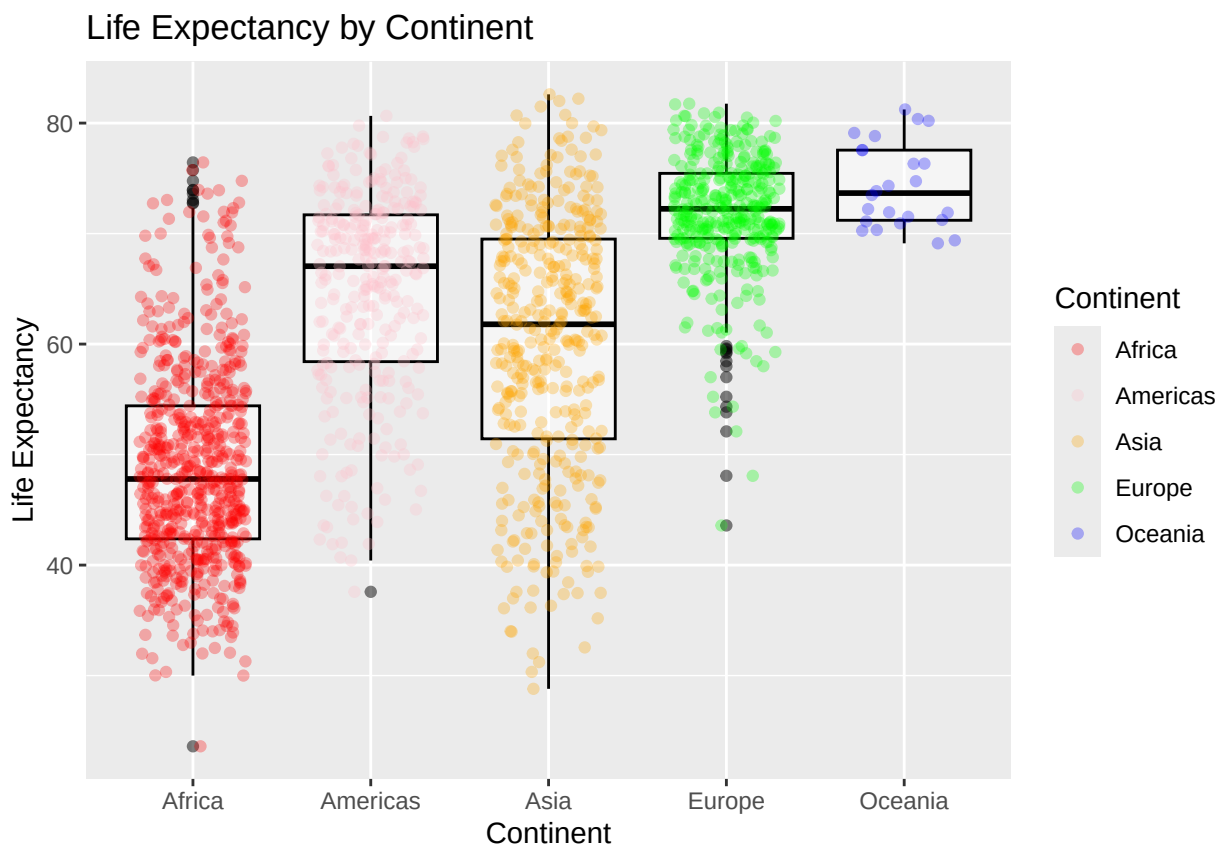
```
#Simple Plot

ggplot(data = gapminder,
            mapping = aes(x = continent, y = lifeExp))+
  geom_point()
```



2. Now, modify your plot! For each change that you make, explain your decision, in terms of the visual elements. **Commit** your changes.

Now for question 2 looking at question one made it hard to understand what the data was showing me, so to make it more clear I made a few changes. First, I created a boxplot to show the median and range of life expectancy. I then made the box plot black so it would be easy to compare with each continent clearly where the range of life expectancy was. I also added jittered points which means that it spreads the points out to prevent them from being as cluttered and overlapping as they were in question 1. Next, I added color to make it easy to differentiate each continent from one another. After doing this, I then assigned each continent a color to make it easy to draw comparisons. Lastly, I improved the labels and gave it a title, x and y axis to help the reader to know how to interpret this data visual.

```
ggplot(gapminder,
       mapping = aes(x = continent, y = lifeExp,
                     col = continent))+
  geom_boxplot(alpha = 0.5, color = "black") +
  geom_jitter(alpha = 0.3, width = 0.3)+
  scale_color_manual(values = c("red", "pink", "orange", "green", "blue"))+
  labs(title = "Life Expectancy by Continent",
       x = "Continent",
       y = "Life Expectancy",
       col = "Continent")
```



3. Use the code provided to load data on racial compositions in California. Then create your own visualization! Try to make it different than what is done above. Is there anything that you want to change, but are unable to? Describe what you want to communicate with these data. **Commit** your changes.

I used the data to show how California's racial composition has changed from 2000-2020. I used what we learned in class to use bar chart to show the various percentages of each ethnic group. By keeping them separated from each time group 2000, 2010, 2020 it allows you to see the difference between each demographic over a large period of time. Before I started, I used head() and names() to understand the data set and organize how I want to convey this information. I made the data into the long format because I thought it would be better to more cleanly showing the data in one column "Year" and the percents into the column "Percent" instead of all in separate one's. Finally, I used facet wrap(~Year) to separate the visual into each year into its own panels to clearly show the change over time, and to make for quick comparison.

```
library(readr)
library(RCurl)
```

```
## Warning: package 'RCurl' was built under R version 4.5.2
```

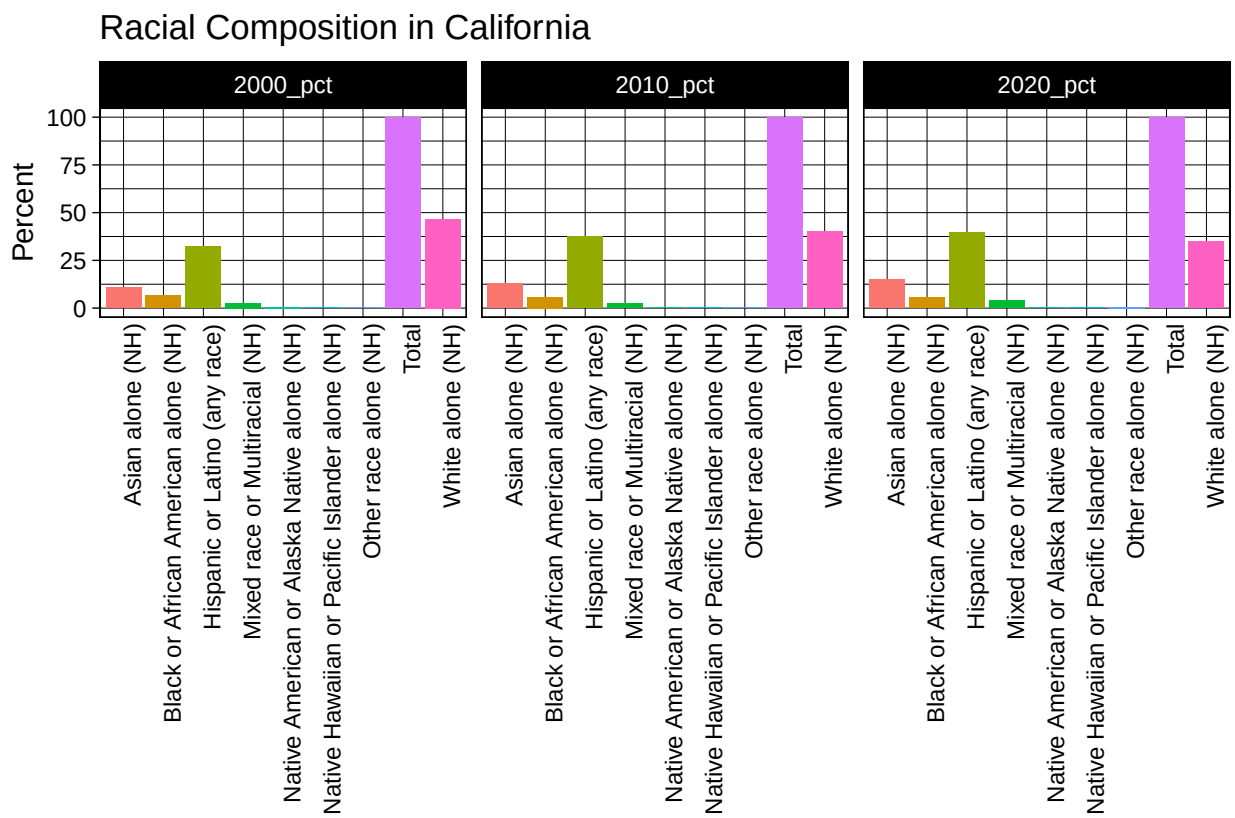
```
# input url from github
url <- getURL("https://raw.githubusercontent.com/tylermcdaniel/soc10/refs/heads/deployment/Data/ca_dem_

# download wikipedia data (from github)
ca_dem_table <- read_csv(url)

#To see the varibles
head(ca_dem_table)
```

```
## # A tibble: 6 x 7
##   Race         `2000_num` `2010_num` `2020_num` `2000_pct` `2010_pct` `2020_pct`
##   <chr>             <dbl>      <dbl>      <dbl>      <dbl>      <dbl>      <dbl>
## 1 White alone~   15816790   14956253   13714587      46.7       40.2       34.7
## 2 Black or Af~    2181926    2163804    2119286       6.44       5.81       5.36
## 3 Native Amer~     178984     162250     156085       0.53       0.44       0.39
## 4 Asian alone~    3648860    4775070    5978795      10.8       12.8       15.1
## 5 Native Hawa~     103736     128577     138167       0.31       0.35       0.35
## 6 Other race ~      71681      85587     223929       0.21       0.23       0.57
```

```r
names(ca_dem_table)
```

```
## [1] "Race"     "2000_num" "2010_num" "2020_num" "2000_pct" "2010_pct" "2020_pct"
```

```r
#Download everything
library(tidyr)
library(magrittr)
library(dplyr)

# Reformat table
ca_dem_table %<>%
  pivot_longer(cols = c("2000_pct", "2010_pct", "2020_pct"),
               names_to = "Year",
               values_to = "Percent")
#Plot data
ggplot(ca_dem_table,
       aes(x = Race, y = Percent, fill = Race)) +
  geom_bar(stat = "identity") +
  labs(title = "Racial Composition in California",
       x = "",
       y = "Percent") +
  theme_linedraw() +
  theme(axis.text.x = element_text(angle = 90, hjust = 1),
        axis.ticks.x = element_blank(),
        legend.position = "none") +
  facet_wrap(~Year)
```

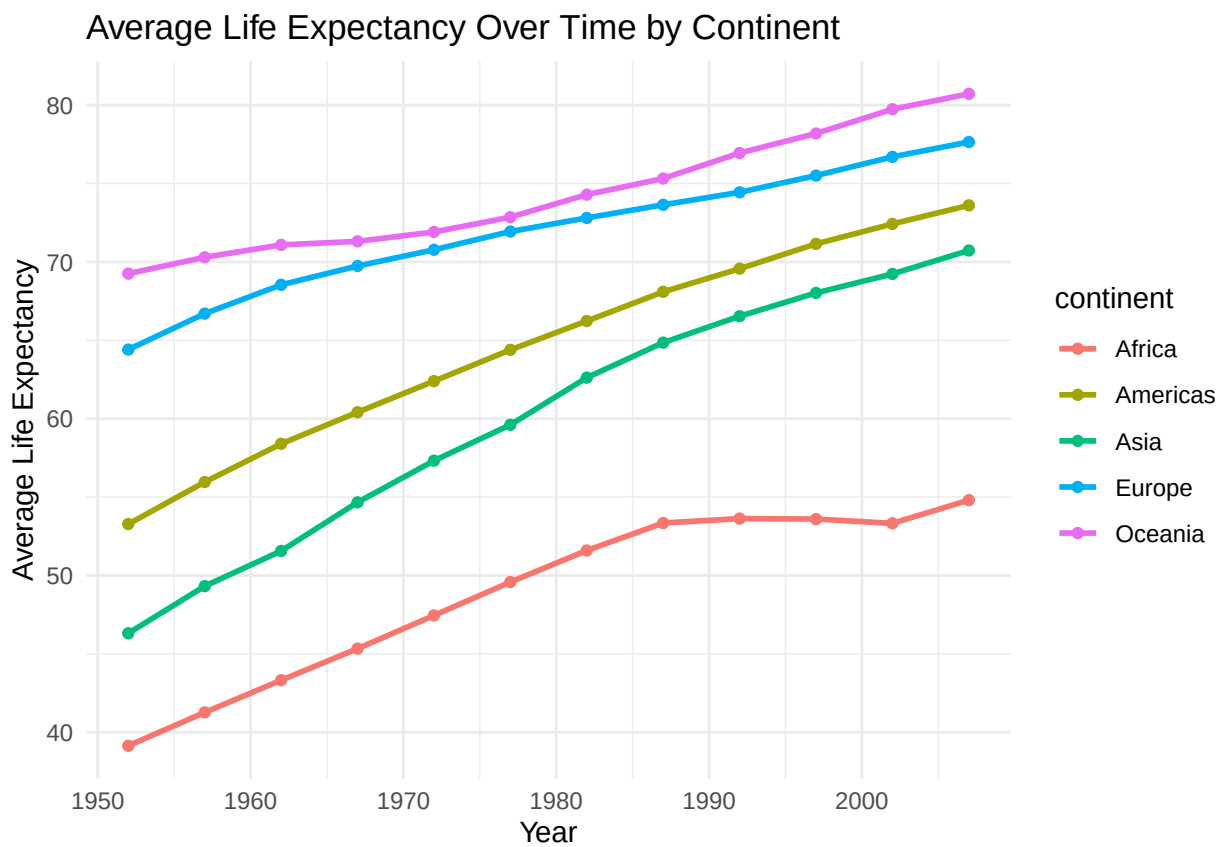


4. Return to a dataset you have worked with in this class already (for example, hand-collected data, data using the `data()` function. The `gtrendsR` data could also work but may need some cleaning/manipulation). Use `ggplot()` to plot this data in a new way. Describe whether the viewer learns anything new from seeing the data in this way. **Commit** your changes.

I used the data set gapminder to find a different way to visualize the changes in average life expectancy over time across each continent. In comparison to the box plot I did in question 1 I think this line plot actually allows you to more quickly understand which continent has the highest average of life and grasp the comparison more efficient. When looking broadly at this graph it is easy to see that each continent has improve slightly in life expectancy, with some more than others but all showing improvement. Overall, I think this graph allowed to see long term trend more clearly to compare to other continents.

```
library(ggplot2)
library(dplyr)
library(gapminder)

gapminder %>%
  group_by(continent, year) %>%
  summarize(avg_lifeExp = mean(lifeExp)) %>%
  ggplot(aes(x = year, y = avg_lifeExp, color = continent)) +
  geom_line(linewidth = 1) +
  geom_point() +
  labs(title = "Average Life Expectancy Over Time by Continent",
       x = "Year",
       y = "Average Life Expectancy") +
  theme_minimal()
```

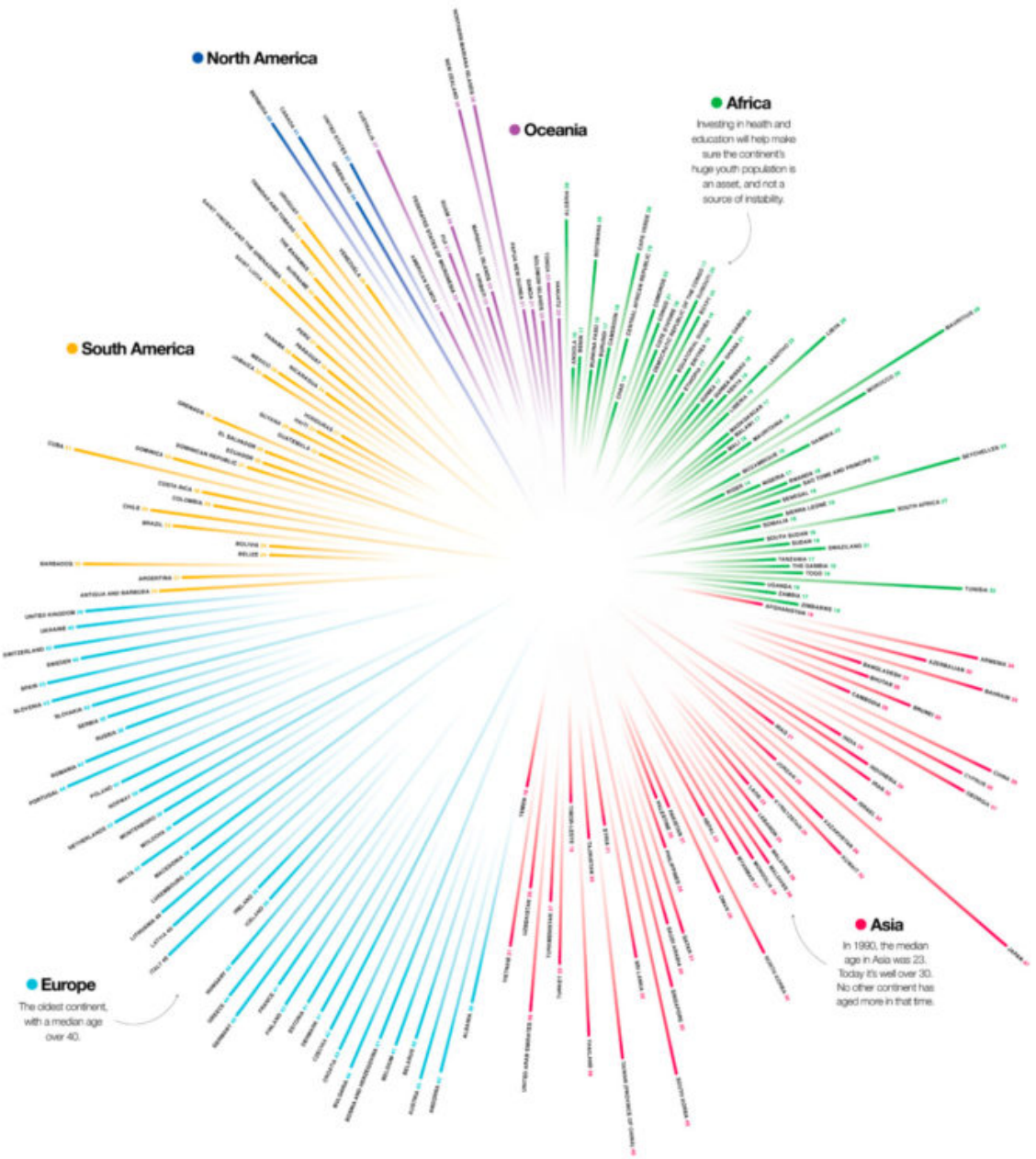
```
## `summarise()` has grouped output by 'continent'. You can override using the
## `.groups` argument.
```



5. Find a data visualization in the wild! It could be in the news, from an academic article, or somewhere else. Include an image of it using the `knitr::include_graphics("")` function. Describe its positive and negative aspects. **Commit** your changes.

The main positive of this graph is that it uses a radial histogram to display a large amount of data in a compact, creative and engaging way. Meaning, compared to a standard bar chart this format allows a lot of values to be shown while clearly separating each group by color. The circular burst makes it easy to see each category, making it simple to see Europe being the highest with the most values extended out and the smallest being Africa with the least values extended out. Hence, the positives are that this graph can convey a lot of data in a small space, as well as using color and this burst effect to allow for easier comparisons to the other continents. However, there are also several negatives to this graph. First, it is hard to determine the exact values for each group because there are no labeled axes or reference points. The white fade spot in the middle also makes it hard to see precisely where the center is accurately. I think this shape also distracts from the information it is trying to convey with the shape and fails to clearly show the exact values and median for each group making it harder to interpret the data. So yes this graph is visually interesting and efficiently using a small space to convey a lot of information but because of that it fails to clearly show the values and how the numerical median varies from continent to continent.

```
knitr::include_graphics("sochw.jpg")
```



Median Age of Population

Africa is the youngest continent, with a median age of just 18. In North America, the median age is 35.

Source: IHME. Data show median age in 2017. Countries grouped by UN regional classifications.